

Graph-Based Generative Design of Mortise-and-Tenon Joints in Timber Structures

Mingli Sun¹, Ying Zhou¹, Yuancao Liu¹, Yuxuan Zhu¹, Qinhui Yang¹, Yige Liu¹, Hua Chai¹

Abstract. Traditional mortise-and-tenon joint design relies heavily on empirical knowledge, struggling to integrate into digital fabrication and circular architecture. This study proposes a unified graph-based paradigm that encodes geometric and topological characteristics into structured data, thereby transforming the design reasoning process into a link prediction task of graph learning. By discretising the joint into the voxel matrix, it quantitatively encodes material configurations and axial kinematic constraints into heterogeneous graph data. A hierarchical extensible standard database comprising 256 two- and three-component joints was established and expanded to a total of 1,536 samples through data augmentation. A parallel hybrid encoder—integrating Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), and Graph SAGE with learnable attention weights—was developed to perform feature fusion. Results demonstrate an AUC-ROC of 0.8571 and an Accuracy of 0.7865. This research facilitates a paradigm shift from experience-based craftsmanship toward a data-driven design paradigm in modern timber engineering.

Keywords: Graph Learning, Mortise-and-Tenon Joint, Integrated Design and Assembly, Generative Design

1 Introduction

Computational design and automated assembly are increasingly prevalent in timber construction. The mortise-and-tenon joint, a connection type in timber engineering, achieves stability solely through the interference of components without adhesives or metal fasteners. This mechanism avoids fibre damage from external processing, enabling easy disassembly and reuse. Compared to conventional metal connectors, mortise-and-tenon joints significantly reduce embodied carbon while

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providing structural self-adaptability through semi-rigid characteristics, offering technical assurance for the secondary high-value utilization of timber (Fang & Mueller, 2023). These properties align with circular construction principles and are vital for the full-life-cycle sustainability of timber architecture.

Traditional mortise-tenon design relies on craft experience and manual reasoning, hindering its integration into contemporary digital fabrication and circular construction. Transitioning from experience-based crafts to generalized data-driven paradigms remains a critical challenge (van Nimwegen & Latteur, 2023). While rule-based parametric modelling enables partial automation (Larsson et al., 2020; Magrisso et al., 2018), it exhibits limited generalizability and scalability. The inherent complexity—stemming from coupled constraints of geometry, topology, structural performance, and assembly (Rogean et al., 2022)—is challenging to capture through explicitly defined rules, especially in multi-component joints.

Deep learning excels at capturing and generalizing latent dependencies within large-scale datasets. Specifically, graph learning offers a specialized trajectory to address the complex structural challenges mentioned above. Graph Neural Networks (GNNs) generalize deep learning to non-Euclidean data via an iterative message-passing mechanism (Scarselli et al., 2009). Unlike traditional models that require rigid data regularity, GNNs utilize iterative aggregation to capture complex interdependencies between nodes. This mechanism ensures permutation invariance and equivariance, providing a robust foundation for subsequent architectures like GCN, GraphSAGE, and GAT.

To facilitate a more generalized design approach for mortise-and-tenon joints, this study proposes a graph-based generative paradigm structured into three phases (Fig.1): graph representation and dataset, graph learning and validation, and application. This paradigm encodes geometric-topological characteristics into structured data, transforming the design reasoning process into a link prediction task in graph learning.

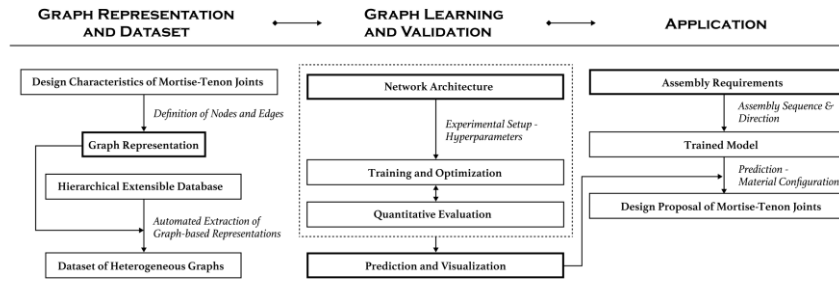


Fig. 1 Graph-based generative design paradigm for mortise-and-tenon joints

2 Graph Representation and Dataset

This section presents the transformation of mortise-and-tenon joint designs into graph representations, the construction of a hierarchical, extensible database, and the generation of a corresponding graph dataset.

2.1 Graph Representation

A mortise-and-tenon joint comprises components and a joint zone, where graph representations encode both material configurations and axial kinetic constraints. For precise quantification, the joint is discretised into a $4 \times 4 \times 4$ voxel matrix (Fig. 2). This 64-voxel resolution captures essential geometric-topological characteristics and aligns with fabrication modules, effectively balancing descriptive precision with computational efficiency.

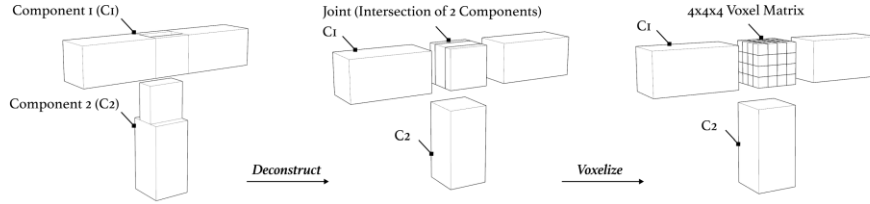


Fig. 2 Voxel-based representation of joint

The study initially defines two categories of nodes: voxel nodes and component nodes (Fig. 3a). The geometric centres of 64 voxels are represented as v_i , while a voxel hypernode v_{hyper} , a non-physical entity, is introduced to facilitate global information flow. Each component is represented by a centroid node C_m^o and six directional nodes ($\pm X, \pm Y, \pm Z$) denoted as $C_m^{+x}, C_m^{-x}, C_m^{+y}$, etc.

Based on these predefined nodes, two types of semantic edges are explicitly defined to represent material configuration and axial kinetic constraints (Fig. 3b). The attribution relationship between voxels and components indicates the material distribution on the joint across different members; for instance, an $e(v_i, C_m^o)$ signifies that v_i is assigned to component m . Axial kinetic constraints are quantified by directed edges from a specific directional node of one component to the centroid node of another, representing kinematic blocking along a particular axis. For example, $e(C_i^{+z}, C_j^o)$ indicates that component i blocks the movement of component j in the $+Z$ direction.

To ensure graph connectivity and stable message propagation, edges are constructed between each component’s centroid and its six directional nodes. The voxel hypernode is linked to all voxel and centroid nodes, facilitating global information flow across the heterogeneous network (Fig. 3b).

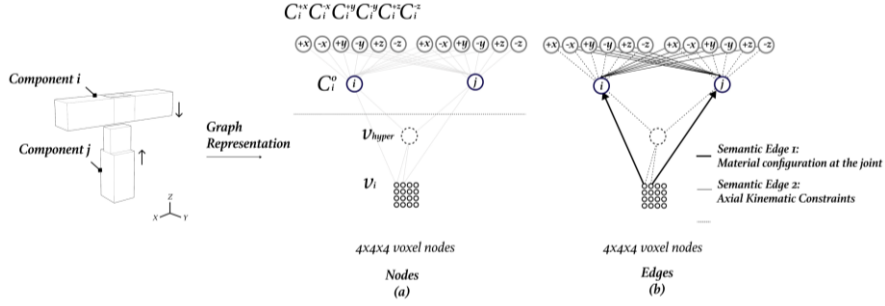


Fig. 3 Definition of nodes and edges

To effectively encode the properties of the mortise-and-tenon joint, node features are configured to ensure informational integrity and topological consistency.

Spatial coordinates ($\mathbf{p}$) define node positions in local or global space. Coordinates for component centroids and the voxel hypernode are set to (0,0,0). Directional nodes use unit vectors $\mathbf{v} \in \{-1, 0, 1\}^3$ for axis-aligned offsets. Voxel nodes are indexed by $(x, y, z) \in \{0, 1, 2, 3\}^3$ within the grid.

Material occupancy ($\mathbf{o}$) is binary-encoded: $o = 1$ if a physical entity exists in a specific direction relative to the joint, and $o = 0$ otherwise. This value is fixed at 0 for centroid nodes and 1 for both voxel nodes and the hypernode.

Component type labels ($\mathbf{y}$) identify specific component entities. For nodes of 2-component and 3-component joints, centroid and directional nodes are assigned labels $y \in \{0, 1, 2\}$ based on their parent components. Voxel nodes and the hypernode are set to a masked state.

2.2 Hierarchical Extensible Standard Database

A comprehensive database of 256 mortise-and-tenon joints was hierarchically organized. The models were divided into 2-member and 3-member joints, and then further refined into four and three subcategories, respectively (Tables 1 and 2). To ensure diversity, each subcategory followed a “reference type + variant” sampling strategy: three distinct reference types were defined for each subcategory, each systematically generating ten variants. This approach produced a well-structured and extensible joint library.

Table 1 Types of 2-component mortise-and-tenon joint

<i>x</i>	<i>x-1</i>	<i>x-2</i>	<i>x-3</i>
			
<i>i</i>	<i>i-1</i>	<i>i-2</i>	<i>i-3</i>
			
<i>l</i>	<i>l-1</i>	<i>l-2</i>	<i>l-3</i>
			
<i>t</i>	<i>t-1</i>	<i>t-2</i>	<i>t-3</i>
			

Table 2 Types of 3-component mortise-and-tenon joint

<i>a</i>	<i>a-1</i>	<i>a-2</i>	<i>a-3</i>
			
<i>b</i>	<i>b-1</i>	<i>b-2</i>	<i>b-3</i>
			
<i>c</i>	<i>c-1</i>	<i>c-2</i>	<i>c-3</i>
			

2.3 Graph Dataset

A procedure was developed to batch-convert 3D geometry into graph data. To augment the dataset, each reference model was rotated by 90° , 180° , and 270° about the x-axis, and by 90° and 270° about the y-axis, generating five rotated variants per sample. This process preserved geometric and topological consistency while updating the numerical features in the graph representation, expanding the dataset from 256 to 1,536 samples. Following data cleaning and parsing, features for Node 1 (component centroid nodes) and Node 2 (voxel nodes) were extracted, and five types of heterogeneous edges were constructed. Each sample was then packaged as a HeteroData object in PyTorch Geometric (PyG) for subsequent graph learning tasks.

3 Model Training and Performance Validation

This section presents a GNN architecture for mortise-and-tenon joint design. Comparative experiments, evaluated across four metrics, demonstrate that the proposed parallel hybrid network outperforms three baseline models in feature representation learning.

3.1 Model Architecture

This study proposes a Parallel Hybrid Graph Neural Network (Parallel Hybrid GNN) architecture for mortise-and-tenon joint design. To address the inconsistent feature dimensions across different node types in the heterogeneous graph, the first layer applies a linear projection to map all node attributes into a unified 32-dimensional latent space. The core architecture consists of three parallel convolutional branches—GCN, GAT, and GraphSAGE—which perform multi-path message aggregation over five types of heterogeneous edges. By enabling three independent branches to aggregate information without mutual interference, the model captures richer high-dimensional node features. An attention fusion module then dynamically integrates the outputs of the three branches, producing node embeddings that better represent high-level geometric-topological features.

The decoder performs link prediction between voxel nodes and component centre nodes, directly learning the material configuration and axial kinematic constraint in joint design. Specifically, the feature embedding of each Node 2 (voxel node) is multiplied via dot product with the feature embeddings of all Node 1 (component centre nodes), yielding a scalar that represents the correlation between

node pairs. This scalar is then transformed into an attribution probability in the range of $[0, 1]$ through a Sigmoid activation function. Finally, the voxel-component assignment is determined by comparing the predicted probabilities of a voxel node across all candidate component categories.

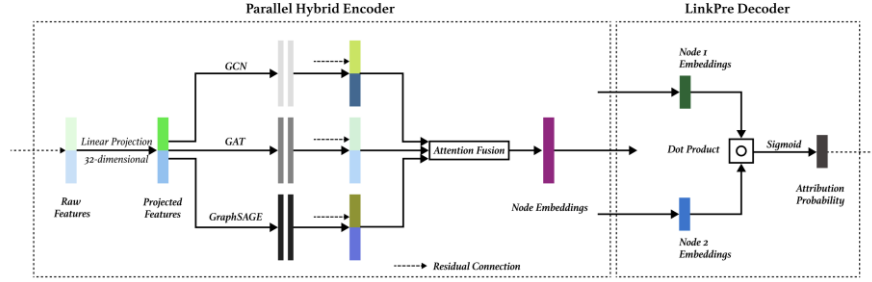


Fig. 4 Architecture of the Parallel Hybrid Graph Neural Network

Weighted Binary Cross-Entropy (WBCE) is utilized as the optimization objective to guide training by minimizing the discrepancy between predicted probabilities and ground-truth labels. By incorporating a weight factor to balance class-wise gradient contributions, this approach enables the Parallel Hybrid GNN to mitigate prediction bias while ensuring stable convergence and high recognition accuracy.

3.2 Experiment Settings

To verify the model's capability in node attribution judgment and the effectiveness of the Parallel Hybrid Graph Neural Network (PHGNN) architecture, this study conducted training sessions under identical parameter settings for four groups of models: the baseline GCN, GAT, and GraphSAGE and the proposed PHGNN architecture. The node attribution performance of these four models was then analysed through various evaluation metrics. The node attribution performance of these four models was then analysed through evaluation metrics.

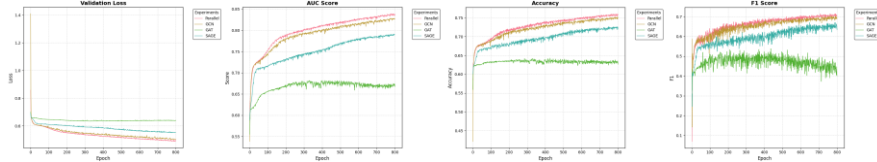


Fig. 5 Comparative training performance and evaluation metrics across different architectures

Training curves demonstrate that the Parallel Hybrid Graph Neural Network significantly outperforms baseline models across all metrics (Fig.5). Over 800 epochs, its ROC-AUC and F1 Score exhibit the steepest growth and highest convergence—with the AUC reaching 0.8571—proving that parallel feature fusion captures complex timber joint topologies more effectively. Compared to standard GCN or GAT architecture, the hybrid approach yields a smoother, lower validation loss and superior accuracy stability, achieving an overall accuracy of 0.7865.

3.3 Visualization and Validation

This study constructed a test set and executed a forward pass using the optimal AUC-performing weights to generate link prediction probabilities. Voxel-based predictions were visualized in 3D space via Matplotlib, employing color-coding to distinguish material distribution across multiple components.

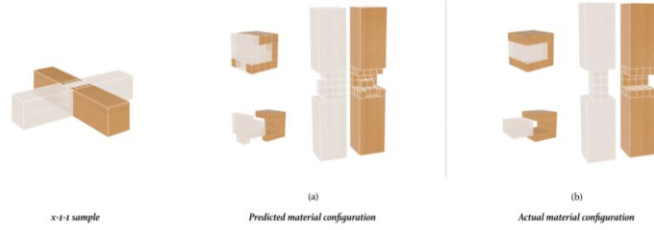


Fig. 6 Comparison between predicted and actual material configuration for sample x-1-1

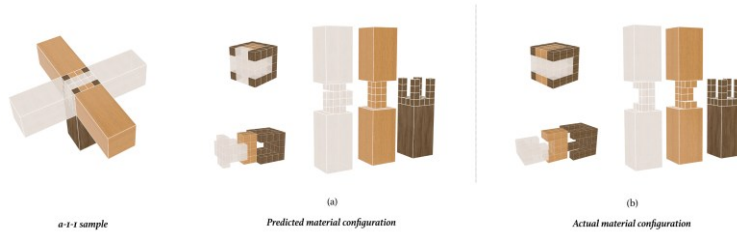


Fig. 7 Comparison between predicted and actual material configuration for sample a-1-1

Fig. 6 and Fig.7 show the comparison between predicted and ground-truth for sample x-1-1 and a-1-1. The data reveals that the predicted structures, like the actual samples, form strictly interlocking joints characterized by kinematic determinacy (a single allowable translation per component). Although discrepancies exist

between the predicted and ground-truth voxel assignments, the axial kinematic constraints between components remain consistent.

4 Application

This study proposes an application framework for the graph-based automated generation of mortise-and-tenon joints. It determines material configurations at each node based on predefined installation sequences and the directions of each component. Timber frame models are established via Rhino-Grasshopper, where an automated script extracts joint topology from the predefined conditions—represented as graph data—and serve as model input. The model then infers the voxel allocation for each joint. Fig. 8 illustrates a preliminary timber design consisting of five components, with predefined installation sequences and directions for each member. Based on the input graph data, the model automatically generates the voxel assignment for joints A and B. This workflow yields high-precision joint solutions under complex constraints.

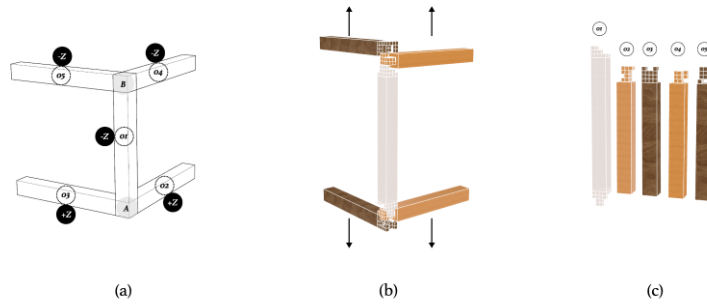


Fig. 8 Design of the mortise-tenon timber frame.

5 Discussion and Conclusion

This study established a unified digital framework using graph learning to transform complex geometric and topological features into non-Euclidean graph data. The proposed Parallel Hybrid GNN effectively captures latent correlations between material configurations and axial kinetic constraints via message passing, achieving an AUC of 0.8571 in the attribution task and demonstrating robust feature-learning capabilities. The proposed framework for application integrates joint design with assembly requirements. By following predefined sequences and orientations, it automatically generates mortise-and-tenon joints satisfying appropriate

material distribution, assembly feasibility, and structural stability—closing the loop from assembly requirements to topological form-finding.

Current limitations include a database of 256 2- and 3-component joints, constraining applicability to complex or non-standard multi-component joints. Additionally, on out-of-dataset data, the model generates voxel assemblies with structural discontinuities. Expanding the database, improving generalization, and refining connectivity will enhance the framework’s robustness and applicability.

Overall, this approach addresses the limitations of experience-based craftsmanship and rule-based parametric modelling, providing a scalable approach for sustainable timber architecture. Aligned with the UN Sustainable Development Goals, this research advances SDGs 11 (Sustainable Cities and Communities) and 12 (Responsible Consumption and Production) through circular construction principles and full-life-cycle sustainability.

6 Acknowledgements

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7 References

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